



Statistics of Financial Markets

Chapter 3: α -stable distributions



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **AIC:** Akaike Information Criterion
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **CLT:** Central Limit Theorem
- ▣ **CMS:** Chambers–Mallows–Stuck (simulation method)
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **ES:** Expected Shortfall
- ▣ **FFT:** Fast Fourier Transform
- ▣ **GCLT:** Generalised Central Limit Theorem
- ▣ **IBM:** International Business Machines (company)
- ▣ **KS:** Kolmogorov–Smirnov (test)
- ▣ **ML:** Maximum Likelihood
- ▣ **QQ:** Quantile–Quantile (plot)
- ▣ **SAS:** Statistical Analysis System (software)
- ▣ **SE:** Standard Error

Today's Question, Route and Learning Outcomes

- **Question:** which probability laws can describe daily returns and also their weekly and monthly sums?
 - ▶ Chapter 2: daily returns have heavier tails than the Normal distribution
 - ▶ log returns add up over time (Chapter 1): the law of a sum matters
- **Route and outcomes:** after this chapter you can
 - ▶ state the stability property and compute the scale of a sum of stable variables
 - ▶ explain the generalised central limit theorem and when sums do not approach the Normal distribution
 - ▶ read the four parameters $(\alpha, \beta, \gamma, \delta)$ and convert between the S0 and S1 parameterisations
 - ▶ say which moments exist for a given α and compare tail probabilities with the Normal distribution
 - ▶ simulate stable variables and estimate their parameters by quantiles and by maximum likelihood
 - ▶ judge, on BET, S&P 500, DAX and Bitcoin returns, where the stable model works and where it fails

Reading and Tools

- Main reference: [Nolan \(2020\)](#), Ch. 1 (definitions, parameterisations, tails) and Ch. 4 (estimation)
 - ▶ short survey with code: [Borak, Härdle & Weron \(2005\)](#); updated in [Borak, Misiorek & Weron \(2011\)](#)
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 3 (probability) and Ch. 18 (heavy tails and extreme risks)
- The original papers: [Mandelbrot \(1963\)](#), [Fama \(1965\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_03](#)
 - ▶ ported from the Quantlet `sim_stable`; each chart links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Stable Distribution](#)

Cotton: an Old Market with Long Price Records

- In the 19th century cotton was one of the most traded commodities in the world
 - ▶ exchanges in New Orleans, New York and Liverpool quoted prices every day
 - ▶ the result: some of the longest daily and monthly price series in economics
- In the early 1960s these series reached Benoit Mandelbrot, then at IBM
 - ▶ he looked at the **changes** of the logarithm of the price, i.e. log returns
- His paper [Mandelbrot \(1963\)](#) started the study of heavy tails in finance



Edgar Degas, A Cotton Office in New Orleans (1873)

Image: Edgar Degas (1873); public domain; Wikimedia Commons

What Mandelbrot Saw

- **Too many large changes:** far more big moves than the Normal distribution allows
 - ▶ the sample variance did not settle as the sample grew
- **The same shape at every horizon:** daily and monthly changes looked alike after rescaling
 - ▶ a sum of many daily changes kept the shape of one daily change
- **His proposal:** stable Paretian laws with $\alpha \approx 1.7$ [Mandelbrot \(1963\)](#)
 - ▶ “Paretian”: the tails decay like a power of x , as in Pareto's law of incomes
 - ▶ consequence: the variance of price changes would be infinite



Benoit Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 fr; [Wikimedia Commons](#)

Fama Tests the Idea on Stocks

- **Fama (1965)**: daily returns of the 30 stocks of the Dow Jones Industrial Average
 - ▶ more large returns than the Normal distribution predicts, for every stock
 - ▶ estimates of α below 2: support for Mandelbrot's hypothesis
- Estimation methods for symmetric stable laws followed
 - ▶ **Fama & Roll (1968)**, **Fama & Roll (1971)**: quantile-based estimates and tables
- Within ten years the evidence started to turn: see the critique at the end of the chapter



Eugene Fama at the Nobel Prize ceremony,
2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0;
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Why the Law of a Sum Matters

- A monthly log return is the sum of about 21 daily log returns
 - ▶ a model for daily returns implies a model for weekly, monthly and yearly returns
- The Normal distribution keeps its shape under summation ([Chapter 2](#))
 - ▶ but it has thin tails: it misses the large daily moves
- **Question for the room:** the Student-t distribution has heavy tails; is a sum of 21 independent Student-t variables again Student-t?
- **Answer**
 - ▶ no: the Student-t family is not closed under summation
 - ▶ if its variance is finite, the sum moves towards the Normal distribution (the central limit theorem)
- **Goal:** find every law whose sums keep the same shape

Definition: Stable Laws

- X is **stable** if, for X_1, X_2 independent copies of X and any $a, b > 0$, there are $c > 0$ and d with
 - ▶ $aX_1 + bX_2 \stackrel{d}{=} cX + d$, where $\stackrel{d}{=}$ means “has the same distribution as”
 - ▶ a, b : the weights of the two copies; c : the scale and d : the shift of the result
 - ▶ **strictly stable** if $d = 0$ for all a, b
- In words: a weighted sum of two copies has the same shape as one copy
 - ▶ only the scale (c) and the location (d) change
- By induction, for n independent copies: $X_1 + \cdots + X_n \stackrel{d}{=} c_n X + d_n$
 - ▶ the only possible scale factor is $c_n = n^{1/\alpha}$ with $0 < \alpha \leq 2$ [Nolan \(2020\)](#)
- α is called the **index of stability** (also: characteristic exponent, tail index)

- Normal: $X_1, X_2 \sim N(0, \sigma^2)$ independent

- ▶ $aX_1 + bX_2 \sim N(0, (a^2 + b^2)\sigma^2)$, so $c = \sqrt{a^2 + b^2}$
- ▶ variances add: $c^2 = a^2 + b^2$, the case $\alpha = 2$

- ▶ $aX_1 + bX_2$ is Cauchy with scale $c = a + b$
- ▶ scales add linearly: $c^1 = a^1 + b^1$, the case $\alpha = 1$
- ▶ the average of n Cauchy draws is again standard Cauchy: averaging does not help



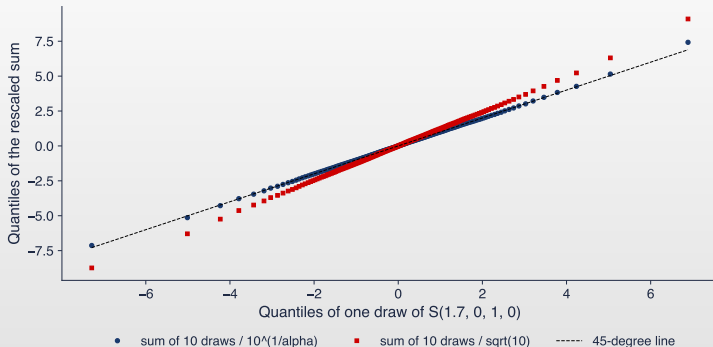
Lithograph: J. Boilly (1821), Wellcome

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The General Rule and a Worked Example

- For X_1, X_2 independent copies of a strictly stable X with index α :
 - ▶ $c^\alpha = a^\alpha + b^\alpha$, i.e. $c = (a^\alpha + b^\alpha)^{1/\alpha}$
- **Example:** $a = 3, b = 4$
 - ▶ $\alpha = 2$: $c = \sqrt{9 + 16} = 5.00$
 - ▶ $\alpha = 1.5$: $c = (3^{1.5} + 4^{1.5})^{1/1.5} = 5.58$
 - ▶ $\alpha = 1$: $c = 3 + 4 = 7.00$
- The smaller α , the larger the scale of the sum: one large term can dominate the sum
- **What do you think?** For $\alpha = 1.5$, what is the scale factor of a sum of two copies?
- **Answer**
 - ▶ $c_2 = 2^{1/1.5} = 2^{2/3} = 1.59$, larger than $\sqrt{2} \approx 1.41$ for the Normal distribution

Stability in a Simulation



- Notation (Section 4): $S(\alpha, \beta, \gamma, \delta)$ is the stable law with index α , skewness β , scale γ and location δ
- Blue: sums of 10 draws of $S(1.7, 0, 1, 0)$ divided by $10^{1/1.7} = 3.87$ lie on the 45-degree line: same law as one draw
- Red: dividing by $\sqrt{10} = 3.16$ (the Normal rule) leaves the tails too wide: 99% quantile 6.31 vs 5.04

Consequences: Horizons and Diversification

- **Horizon:** if daily returns are i.i.d. (independent, identically distributed) $S(\alpha, 0, \gamma, 0)$, the 20-day sum has scale $20^{1/\alpha}\gamma$
 - ▶ $\alpha = 2$: 4.47γ (the square-root-of-time rule); $\alpha = 1.7$: 5.83γ ; $\alpha = 1.5$: 7.37γ
 - ▶ with $\alpha < 2$, the risk of long horizons grows faster than $\sqrt{n}$
- **Diversification:** an equally weighted portfolio of N i.i.d. stable assets has scale $N^{1/\alpha-1}\gamma$
 - ▶ $N = 10$: $\alpha = 2$ gives 0.32γ ; $\alpha = 1.5$ gives 0.46γ ; $\alpha = 1$ gives 1.00γ
 - ▶ with $\alpha = 1$ diversification does not reduce the scale at all; [Fama \(1965\)](#) discussed this point

Recap: Stability

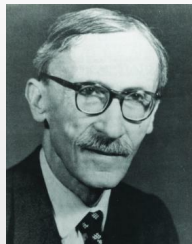
- Stable law: a sum of independent copies has the same shape, only rescaled and shifted
- Scale rule: $c^\alpha = a^\alpha + b^\alpha$; n copies: factor $n^{1/\alpha}$, $0 < \alpha \leq 2$
- Normal: $\alpha = 2$; Cauchy: $\alpha = 1$
- Small α : risk grows faster with the horizon and diversification helps less

From the Classical to the Generalised CLT

- **Classical CLT** (central limit theorem, [Chapter 2](#)): X_i i.i.d. with mean μ and finite variance σ^2
 - ▶ $\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \rightarrow N(0, 1)$ in distribution
- What if the variance is infinite? The normalisation $\sqrt{n}$ is then wrong
- **GCLT** (generalised central limit theorem) [Gnedenko & Kolmogorov \(1954\)](#): if $(X_1 + \dots + X_n - b_n)/a_n$ converges to a non-degenerate law, that law is stable
 - ▶ $a_n > 0$, b_n : normalising constants that rescale and recentre the sum (for the classical CLT, $a_n = \sigma\sqrt{n}$, $b_n = n\mu$)
 - ▶ stable laws are the **only** possible limits of normalised sums of i.i.d. variables
 - ▶ the Normal distribution is the special case $\alpha = 2$

Domains of Attraction

- ▣ The **domain of attraction** of a stable law: all laws whose normalised sums converge to it
 - ▶ finite variance: domain of attraction of the Normal distribution, $a_n \propto \sqrt{n}$
 - ▶ power-law tails $P(|X| > x) \approx Cx^{-\alpha}$ with $0 < \alpha < 2$: the stable law with the same α , $a_n \propto n^{1/\alpha}$
 - ▶ $C > 0$: a constant; $\propto$: proportional to
- ▣ So the **tail** of one return decides the law of long sums
- ▣ Paul Lévy characterised the stable laws in the 1920s; the full theory is in Gnedenko & Kolmogorov (1954)



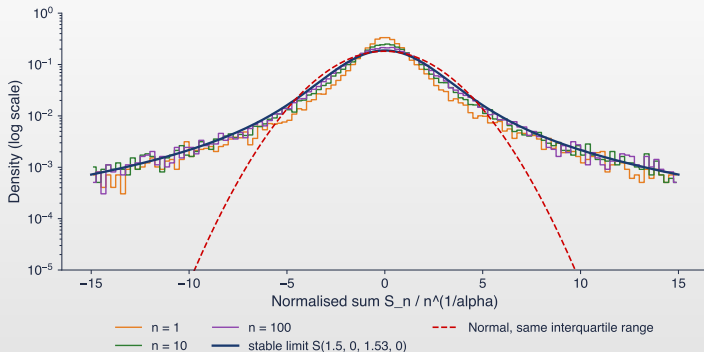
Paul Lévy (1886–1971)

Photo: Konrad Jacobs, Oberwolfach Photo

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[Commons](#)

The GCLT in a Simulation



- Student-t with $\nu = 1.5$ degrees of freedom: infinite variance, tails $\propto x^{-1.5}$; sums of n draws divided by $n^{1/1.5}$
- $n = 1, 10, 100$: the histograms approach the stable limit $S(1.5, 0, 1.53, 0)$; $P(|\cdot| > 10)$: 2.73% for $n = 100$, 2.64% in the limit
- A Normal law with the same interquartile range gives 0.0005%: it misses the tails completely

What the GCLT Means for Returns

- Two possible worlds for daily returns
 - ▶ **tail index below 2** (infinite variance): monthly returns are stable with the **same** α
 - ▶ **tail index above 2** (finite variance): monthly returns move towards the Normal distribution
- The two worlds give different testable predictions about returns at longer horizons
- **Question for the room:** in which world do the S&P 500 returns live?
- **Answer**
 - ▶ we test it with real data in the last sections: estimates of α at daily, weekly and monthly horizons

Recap: The Generalised CLT

- ▣ Classical CLT: finite variance, normalisation $\sqrt{n}$, Normal limit
- ▣ GCLT: stable laws are the only limits of normalised sums
- ▣ Power-law tails with index $\alpha < 2$: normalisation $n^{1/\alpha}$, stable limit with the same α
- ▣ The tail of a single return decides the law of long sums

The Characteristic Function

- **Characteristic function** of X : $\varphi_X(t) = E[e^{itX}] = E[\cos(tX)] + i E[\sin(tX)]$, $t \in \mathbb{R}$
 - ▶ $i = \sqrt{-1}$: the imaginary unit; t : a real argument, not time
 - ▶ it always exists, because $|e^{itX}| = 1$, even when no moment exists
 - ▶ it determines the distribution uniquely
- Two properties used in this chapter
 - ▶ independent X, Y : $\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t)$: sums become products
 - ▶ $\varphi_{aX+b}(t) = e^{ibt} \varphi_X(at)$
- Normal $N(\mu, \sigma^2)$: $\varphi(t) = \exp(i\mu t - \sigma^2 t^2/2)$
- Why we need it: apart from three cases, stable laws have **no closed-form density**; they are defined by φ

The Stable Characteristic Function (S1)

□ $X \sim S(\alpha, \beta, \gamma, \delta; 1)$ if, for $\alpha \neq 1$, Samorodnitsky & Taqqu (1994), Nolan (2020):

▶ $\varphi(t) = \exp \left\{ -\gamma^\alpha |t|^\alpha \left[1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2} \right] + i\delta t \right\}$

▶ for $\alpha = 1$: $\varphi(t) = \exp \left\{ -\gamma |t| \left[1 + i\beta \frac{2}{\pi} \operatorname{sign}(t) \ln |t| \right] + i\delta t \right\}$

□ The four parameters

▶ $\alpha \in (0, 2]$: **index of stability**; smaller α , heavier tails

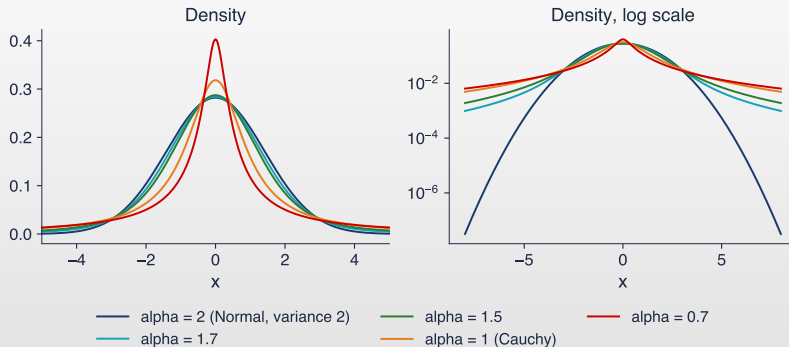
▶ $\beta \in [-1, 1]$: **skewness**; $\beta > 0$ heavier right tail, $\beta < 0$ heavier left tail

▶ $\gamma > 0$: **scale** (it plays the role of σ); $\delta \in \mathbb{R}$: **location**

□ $\operatorname{sign}(t)$: +1 for $t > 0$, -1 for $t < 0$; the term with β makes the law asymmetric

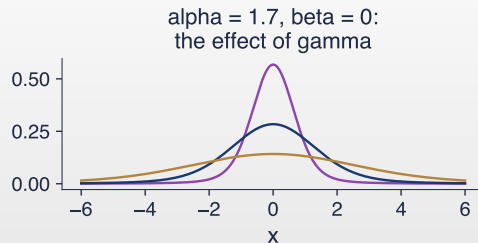
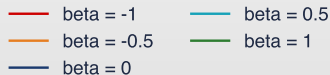
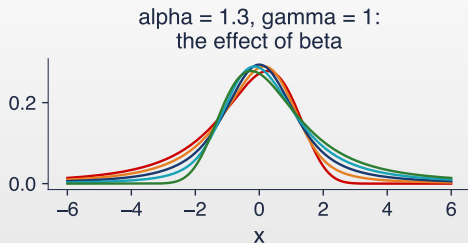
□ Stability is visible: $\varphi(t)^n$ has the same form, with γ^α replaced by $n\gamma^\alpha$

The Effect of α



- Symmetric laws $S(\alpha, 0, 1, 0)$: smaller α gives a taller, narrower peak and much heavier tails
- Log scale: the Normal tail ($\alpha = 2$) falls like $e^{-x^2/4}$; the others fall like a power of x

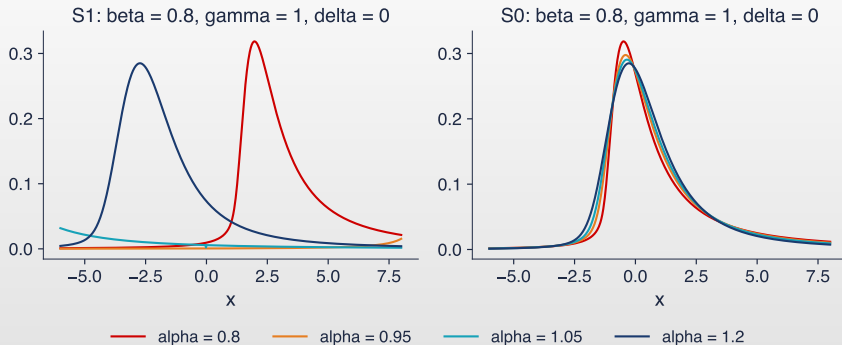
The Effect of β and γ



- β moves weight between the tails: $\beta = 1$ (green) has the heavier right tail, $\beta = -1$ (red) the heavier left tail
- γ stretches the density: doubling γ doubles every quantile around δ

Two Parameterisations: S0 and S1

- ▣ **S0** (Nolan's recommendation for data analysis), $\alpha \neq 1$:
 - ▶ $\varphi(t) = \exp \left\{ -\gamma^\alpha |t|^\alpha \left[1 + i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2} (|\gamma t|^{1-\alpha} - 1) \right] + i\delta_0 t \right\}$
- ▣ Only the location differs: $\delta_0 = \delta_1 + \beta\gamma \tan \frac{\pi\alpha}{2}$ ($\alpha \neq 1$)
 - ▶ δ_0, δ_1 : the location parameter in S0 and in S1; α, β and γ are the same in both
 - ▶ for $\beta = 0$ the two coincide
- ▣ Why two?
 - ▶ S1: simpler algebra for sums; used in most theory books [Samorodnitsky & Taqqu \(1994\)](#)
 - ▶ S0: the density changes continuously with α and β ; better for estimation and plots [Nolan \(2020\)](#)

S1 vs S0 near $\alpha = 1$ 

- S1 (left): with $\beta = 0.8$, the density runs away to $+\infty$ as $\alpha \uparrow 1$ and comes back from $-\infty$ as $\alpha \downarrow 1$, because $\tan \frac{\pi \alpha}{2}$ explodes
- S0 (right): the same four laws, shifted; the mode stays near 0 and changes smoothly

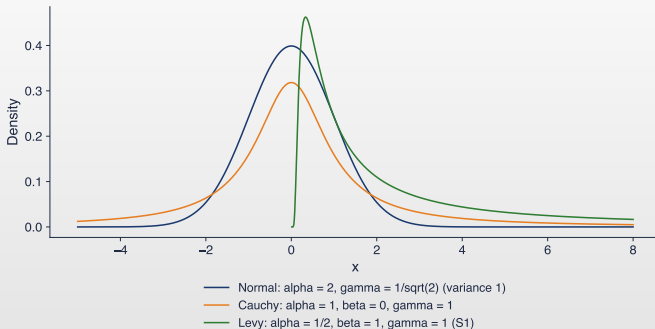
Mind the Parameterisation in Software

- `scipy.stats.levy_stable` uses **S1** by default
 - ▶ switch with `levy_stable.parameterization = 'S0'`
 - ▶ arguments: `levy_stable.pdf(x, alpha, beta, loc=delta, scale=gamma)`
- Papers and programs report δ in S0, in S1 or in other parameterisations: check which one before comparing
- Scale trap: $S(2, 0, \gamma, \delta)$ is $N(\delta, 2\gamma^2)$, not $N(\delta, \gamma^2)$
 - ▶ so $\gamma = \sigma/\sqrt{2}$ for the Normal distribution
- In this course: estimates in S0, with δ_1 also reported

Three Laws with a Closed-Form Density

- **Normal:** $\alpha = 2$ (β plays no role)
 - ▶ $\varphi(t) = \exp(-\gamma^2 t^2 + i\delta t)$, i.e. $N(\delta, 2\gamma^2)$
- **Cauchy:** $\alpha = 1$, $\beta = 0$
 - ▶ $f(x) = \frac{\gamma}{\pi(\gamma^2 + (x - \delta)^2)}$; $\varphi(t) = \exp(-\gamma|t| + i\delta t)$
- **Lévy:** $\alpha = 1/2$, $\beta = 1$ (S1); totally skewed to the right
 - ▶ $f(x) = \sqrt{\frac{\gamma}{2\pi}} (x - \delta)^{-3/2} \exp\left(-\frac{\gamma}{2(x - \delta)}\right)$ for $x > \delta$
 - ▶ the first time a Brownian motion hits a level has a Lévy law
- All other stable densities are computed numerically from φ [Nolan \(1997\)](#), [Mittnik, Doganoglu & Chenyao \(1999\)](#)

Normal, Cauchy and Lévy



- Cauchy: lower peak, much heavier tails than the Normal distribution with variance 1
- Lévy: only positive values, a long right tail; its mean is infinite

Worked Example: S1 vs S0 at $t = 1$

- X with $\alpha = 1.5$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$; note $\tan \frac{3\pi}{4} = -1$
- **S1:** $\ln \varphi(1) = -1 \cdot [1 - i \cdot 0.3 \cdot (-1)] = -1 - 0.3i$
 - ▶ $\varphi(1) = e^{-1}(\cos 0.3 - i \sin 0.3) = 0.351 - 0.109i$; $|\varphi(1)| = e^{-1} = 0.368$
- **S0:** the factor $|t|^{1-\alpha} - 1$ is 0 at $t = 1$, so $\varphi(1) = e^{-1} = 0.368$, a real number
 - ▶ same modulus: the two laws differ only by a shift
 - ▶ shift $= \beta\gamma \tan \frac{\pi\alpha}{2} = -0.3$: the S1 law with $\delta_1 = 0$ is the S0 law with $\delta_0 = -0.3$

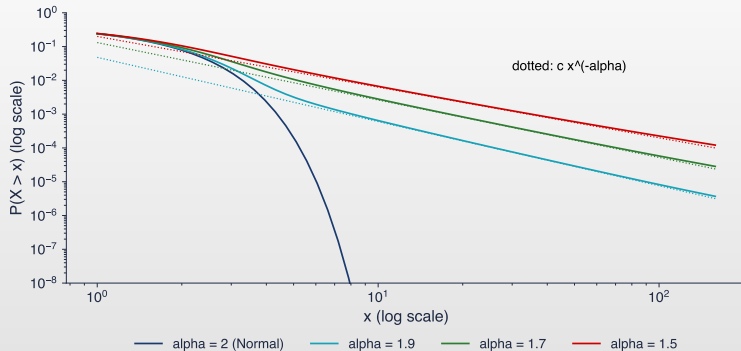
Recap: Parameters and the Characteristic Function

- ▣ Stable laws are defined by $\varphi(t)$; the density is computed numerically
- ▣ α tails, β skewness, γ scale, δ location
- ▣ S0 vs S1: only δ differs, $\delta_0 = \delta_1 + \beta\gamma \tan \frac{\pi\alpha}{2}$; scipy uses S1 by default
- ▣ Closed forms: Normal ($\alpha = 2$, variance $2\gamma^2$), Cauchy, Lévy

Power-Law Tails

- For $\alpha < 2$, as $x \rightarrow \infty$ [Nolan \(2020\)](#):
 - ▶ $P(X > x) \sim \gamma^\alpha c_\alpha (1 + \beta) x^{-\alpha}$ and $P(X < -x) \sim \gamma^\alpha c_\alpha (1 - \beta) x^{-\alpha}$
 - ▶ $c_\alpha = \Gamma(\alpha) \sin(\pi\alpha/2)/\pi$, with Γ the gamma function; $f \sim g$ means $f/g \rightarrow 1$
- On log-log axes the tail is a straight line with slope $-\alpha$
 - ▶ doubling the threshold divides the tail probability by 2^α : $\times 0.31$ for $\alpha = 1.7$
- The Normal tail falls like $e^{-x^2/(4\gamma^2)}$: doubling the threshold makes it vanish
- $\beta = \pm 1$ with $\alpha < 1$: one tail is missing (the law lives on a half-line)

Tails on Log-Log Axes

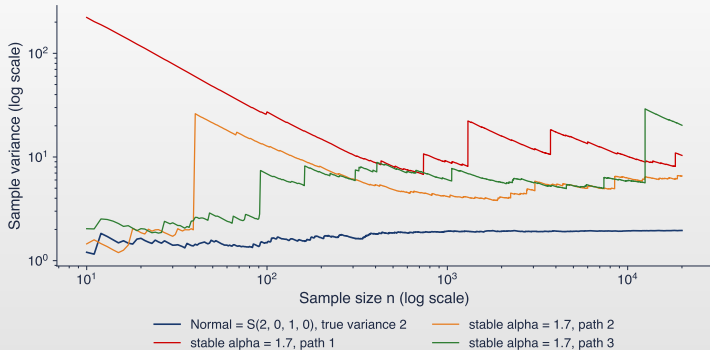


- Solid: $P(X > x)$ of $S(\alpha, 0, 1, 0)$; dotted: the power law $c_\alpha x^{-\alpha}$, reached already near $x = 5$
- $P(X > 10)$: 0.279% for $\alpha = 1.7$, 0.667% for $\alpha = 1.5$, about $8 \times 10^{-11}\%$ for the Normal distribution

Which Moments Exist?

- For $0 < \alpha < 2$: $E|X|^p < \infty$ if and only if $p < \alpha$
 - ▶ $E|X|^p$: the absolute moment of order $p > 0$; $p = 1$ gives the mean, $p = 2$ the variance
 - ▶ reason: $|x|^p$ times a density of order $|x|^{-\alpha-1}$ is integrable only if $p < \alpha$
- Consequences
 - ▶ $\alpha < 2$: **infinite variance**; skewness and kurtosis are not defined
 - ▶ $1 < \alpha < 2$: the mean exists; in S1 it equals δ_1
 - ▶ $\alpha \leq 1$: not even the mean exists (Cauchy, Lévy)
 - ▶ $\alpha = 2$: all moments exist
- **What do you think?** Mandelbrot found $\alpha \approx 1.7$ for cotton. Does the mean daily change exist?
- **Answer**
 - ▶ yes, because $1 < 1.7$; but the variance does not, because $2 > 1.7$

The Sample Variance Does Not Settle



- Normal (blue): the sample variance converges to the true value 2
- Stable $\alpha = 1.7$: each large draw makes the variance jump; after 20 000 draws the three paths end at 10.4, 6.6, 20.2
- Mandelbrot saw exactly this in cotton prices

Tail Probabilities: Stable vs Normal

k	$P(X > k), \text{Normal } \%$	$P(X > k), \alpha = 1.7 \%$	ratio
2	15.73	18.59	1.2
3	3.39	7.25	2.1
5	0.041	2.13	52.4
10	1.5×10^{-10}	0.56	3.6×10^9

- Both laws with $\beta = 0, \gamma = 1, \delta = 0$: the Normal law is $N(0, 2)$
- Near the centre the two laws are close; far in the tails they differ by orders of magnitude
- A risk model is judged in the tails, where the choice of law matters most

Infinite Variance and Risk Measures

- Tools that need a finite variance
 - ▶ volatility, the Sharpe ratio and its standard error ([Chapter 1](#)); mean–variance portfolios; confidence intervals based on the CLT
 - ▶ with $\alpha < 2$ these numbers exist in every sample but estimate nothing
- Tools that survive
 - ▶ quantiles always exist: $\text{VaR}_{1\%} = -q_{1\%}$, the loss exceeded with probability 1%
 - ▶ $\text{ES}_{2.5\%}$ (expected shortfall, the mean loss beyond the VaR) exists only if $\alpha > 1$
 - ▶ the scale γ and the interquartile range replace the standard deviation
- VaR and ES are developed in [Chapter 10](#)

Recap: Tails and Moments

- ▣ Tails: $P(|X| > x) \propto x^{-\alpha}$; a straight line on log-log axes
- ▣ Moments: $E|X|^p < \infty$ iff $p < \alpha$; $\alpha < 2$ means infinite variance
- ▣ The sample variance of a stable sample jumps and does not settle
- ▣ Quantiles, VaR and the scale γ remain meaningful

The Chambers–Mallows–Stuck Method (Symmetric Case)

- No closed-form density, yet stable variables are easy to simulate [Chambers, Mallows & Stuck \(1976\)](#)
- **CMS** (Chambers–Mallows–Stuck) for $S(\alpha, 0, 1, 0)$:
 - ▶ draw $V \sim U(-\pi/2, \pi/2)$, a uniform angle, and $W \sim \text{Exp}(1)$, an exponential variable with mean 1, independent
 - ▶ $X = \frac{\sin(\alpha V)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1-\alpha)V)}{W} \right)^{(1-\alpha)/\alpha}$
 - ▶ then $\gamma X + \delta \sim S(\alpha, 0, \gamma, \delta)$
- Checks on special cases
 - ▶ $\alpha = 1$: $X = \tan V$, a Cauchy variable
 - ▶ $\alpha = 2$: $X = 2 \sin V \sqrt{W}$, a $N(0, 2)$ variable

The General Case ($\beta \neq 0$)

- For $\alpha \neq 1$, S1 parameterisation [Weron \(1996\)](#):

$$\blacktriangleright B = \frac{\arctan(\beta \tan \frac{\pi\alpha}{2})}{\alpha}, \quad S = \left(1 + \beta^2 \tan^2 \frac{\pi\alpha}{2}\right)^{1/(2\alpha)}$$

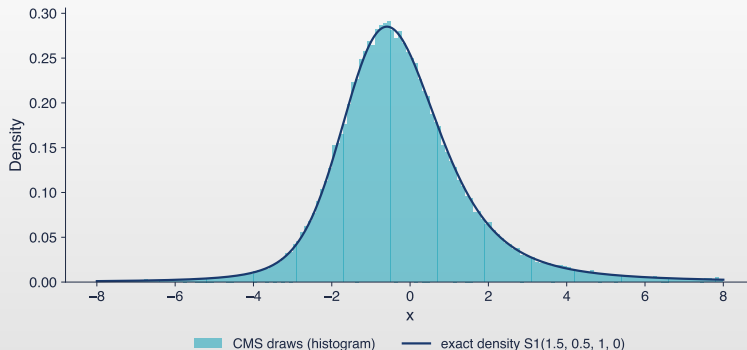
$$\blacktriangleright X = S \frac{\sin(\alpha(V + B))}{(\cos V)^{1/\alpha}} \left(\frac{\cos(V - \alpha(V + B))}{W} \right)^{(1-\alpha)/\alpha} \sim S(\alpha, \beta, 1, 0; 1)$$

- B, S : auxiliary constants that depend only on α and β ; for $\beta = 0$, $B = 0$ and $S = 1$
- From the standard draw to any parameters
 - $S1$: $\gamma X + \delta_1$; $S0$: $\gamma X + \delta_0 - \beta \gamma \tan \frac{\pi\alpha}{2}$
- [Weron \(1996\)](#) corrected a formula of the original paper for $\beta \neq 0$; `scipy.stats.levy_stable.rvs` uses the same method

Worked Example: One Draw Step by Step

- ▣ $\alpha = 1.5$, $\beta = 0$; a uniform draw $U = 0.8$ and an exponential draw $W = 1.2$
- ▣ Steps
 - ▶ $V = \pi(U - 1/2) = 0.942$: turns U , uniform on $(0, 1)$, into a uniform angle on $(-\pi/2, \pi/2)$
 - ▶ $\sin(\alpha V) = 0.988$; $(\cos V)^{1/\alpha} = 0.702$
 - ▶ $(\cos((1 - \alpha)V)/W)^{(1-\alpha)/\alpha} = 1.104$
 - ▶ $X = 0.988/0.702 \times 1.104 = 1.554$
- ▣ Repeat with new (U, W) pairs to obtain a sample of any size

CMS Draws against the Exact Density



- 100 000 CMS draws of $S(1.5, 0.5, 1, 0; 1)$: the histogram follows the density computed from φ
- Quantiles 1% and 99%: sample -5.47 and 9.52 , exact -5.40 and 9.76
- Two-sample KS (Kolmogorov–Smirnov) test against `levy_stable.rvs`: statistic 0.0044 , p-value 0.30 , no difference

Recap: Simulation

- CMS: a uniform angle V and an exponential W give one stable draw
- Special cases check the formula: $\tan V$ (Cauchy), $2 \sin V \sqrt{W}$ (Normal)
- Skewed case: the formula of [Weron \(1996\)](#); shift by $\beta\gamma \tan \frac{\pi\alpha}{2}$ for S_0

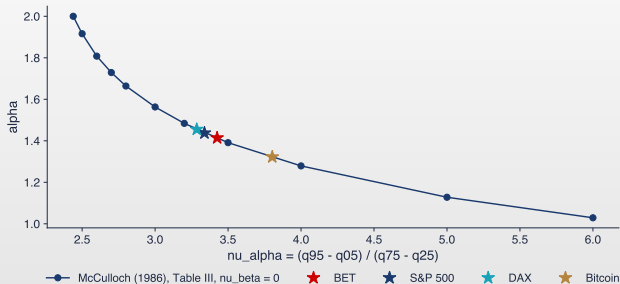
Three Families of Estimators

- ▣ **Quantile methods:** match sample quantiles to those of the stable law
 - ▶ Fama & Roll (1971) for symmetric laws; McCulloch (1986) for all four parameters
- ▣ **Characteristic-function methods:** regress the empirical $\ln(-\ln|\hat{\varphi}(t)|^2)$ on $\ln|t|$
 - ▶ $\hat{\varphi}(t) = \frac{1}{n} \sum_{j=1}^n e^{itr_j}$: the empirical characteristic function; $|\varphi(t)|^2 = e^{-2\gamma^\alpha|t|^\alpha}$, so the log-log relation is linear in $\ln|t|$
 - ▶ Koutrouvelis (1980); the slope estimates α
- ▣ **Maximum likelihood (ML):** the most precise, once the density can be computed
 - ▶ Nolan (2001), Mittnik et al. (1999)
- ▣ A SAS implementation of these estimators is described in Pele (2014)

McCulloch's Quantile Method

- Two ratios of sample quantiles $\hat{q}_p$, free of location and scale [McCulloch \(1986\)](#):
 - ▶ $\nu_\alpha = \frac{\hat{q}_{0.95} - \hat{q}_{0.05}}{\hat{q}_{0.75} - \hat{q}_{0.25}}$ (tail width relative to the body)
 - ▶ $\nu_\beta = \frac{\hat{q}_{0.95} + \hat{q}_{0.05} - 2\hat{q}_{0.5}}{\hat{q}_{0.95} - \hat{q}_{0.05}}$ (asymmetry)
- $\hat{q}_p$: the sample quantile of order p , the value below which a share p of the returns lies
- Tables III and IV of the paper map (ν_α, ν_β) to $(\hat{\alpha}, \hat{\beta})$
 - ▶ Normal distribution: $\nu_\alpha = 2.439$; larger ν_α means smaller α
- Then $\hat{\gamma} = (\hat{q}_{0.75} - \hat{q}_{0.25})/\phi_3(\hat{\alpha}, \hat{\beta})$ (Table V) and $\hat{\delta}_0$ from Table VII; ϕ_3 : the interquartile range of $S(\alpha, \beta, 1, 0)$
- Fast, consistent, valid for $0.6 \leq \alpha \leq 2$; less precise than ML; a good starting point for ML

McCulloch's Map from ν_α to α



- Line: Table III for a symmetric sample ($\nu_\beta = 0$)
- Stars: daily log returns since 2000 (Bitcoin since 2014)
- ν_α : BET 3.425, S&P 500 3.338, DAX 3.285, Bitcoin 3.803
- All well above 2.439: tails much wider than the Normal distribution allows

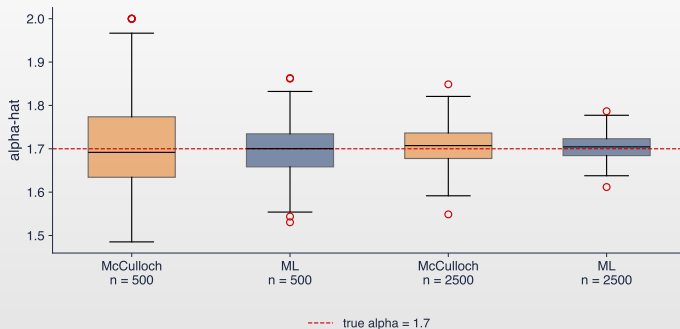
Worked Example: McCulloch on the BET

- ▣ BET daily log returns (%), 2000–2026, $n = 6\,670$
- ▣ Step 1: quantiles
 - ▶ $\hat{q}_{0.05} = -1.93$, $\hat{q}_{0.25} = -0.50$, $\hat{q}_{0.5} = 0.08$, $\hat{q}_{0.75} = 0.66$, $\hat{q}_{0.95} = 2.06$
- ▣ Step 2: ratios
 - ▶ $\nu_\alpha = 4.00/1.17 = 3.425$; $\nu_\beta = -0.006$, practically symmetric
- ▣ Step 3: Table III, column $\nu_\beta = 0$: $\nu_\alpha = 3.2 \mapsto 1.484$, $\nu_\alpha = 3.5 \mapsto 1.391$
 - ▶ linear interpolation: $\hat{\alpha} \approx 1.414$
- ▣ Step 4: $\hat{\gamma} = 0.60$ (% a day), $\hat{\beta} = -0.011$, $\hat{\delta}_0 = 0.079$

Maximum Likelihood

- Log-likelihood of a sample $r_1, \dots, r_n$: $\ell(\theta) = \sum_{t=1}^n \ln f(r_t; \theta)$, $\theta = (\alpha, \beta, \gamma, \delta_0)$
 - ▶ θ : the vector of the four parameters; $f(r_t; \theta)$: the stable density at r_t ; $\hat{\theta}$ maximises ℓ numerically, starting from the McCulloch estimate
 - ▶ standard errors from the curvature of ℓ : the inverse of the Hessian, the matrix of second derivatives of ℓ
- The density f is computed from φ
 - ▶ $f(x) = \frac{1}{2\pi} \int e^{-itx} \varphi(t) dt$, evaluated on a grid with the FFT (fast Fourier transform) [Mittnik, Doganoglu & Chenyao \(1999\)](#)
 - ▶ or by direct integration [Nolan \(1997\)](#); `levy_stable.fit` does this and needs minutes for a few thousand returns
- ML in S0 is regular for $\alpha \in (0, 2)$: standard errors shrink like $1/\sqrt{n}$ [Nolan \(2001\)](#)

McCulloch vs ML in a Monte Carlo Study



- 200 samples of $S(1.7, 0, 1, 0)$ for each size; both estimators are centred on 1.7
- Standard deviation of $\hat{\alpha}$: $n = 500$: McCulloch 0.107, ML 0.061; $n = 2500$: 0.046 vs 0.030
- ML is about 1.7 times more precise for the same data

Recap: Estimation

- ▣ McCulloch: five quantiles, two ratios, four tables; fast and robust
- ▣ ML: numerical density from φ , numerical maximisation; the most precise
- ▣ Report the parameterisation (S0 or S1) and the units of γ and δ

Stable Fits of Daily Log Returns

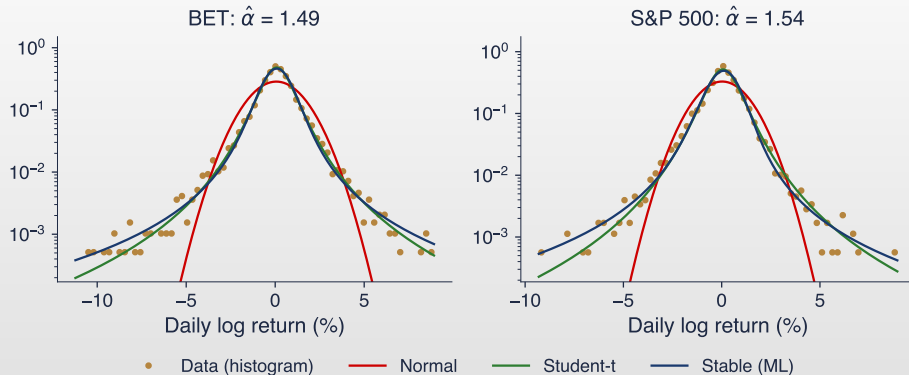
	n	$\hat{\alpha}$ McC.	$\hat{\alpha}$ ML (SE)	$\hat{\beta}$ ML (SE)	$\hat{\gamma}$	$\hat{\delta}_0$	$\hat{\delta}_1$
BET	6 670	1.414	1.486 (0.020)	−0.016 (0.034)	0.62	0.088	0.078
S&P 500	6 714	1.438	1.540 (0.021)	−0.181 (0.036)	0.58	0.093	0.000
DAX	6 786	1.455	1.606 (0.021)	−0.179 (0.040)	0.74	0.090	−0.004
Bitcoin	4 384	1.322	1.410 (0.026)	0.019 (0.036)	1.54	0.132	0.171

- Daily log returns in %; indices 2000–2026, Bitcoin 2014–2026 (7 days a week); data: EODHD
- McC.: McCulloch; ML in S0 with standard errors; $\hat{\gamma}$, $\hat{\delta}_0$, $\hat{\delta}_1$ in % a day

Reading the Estimates

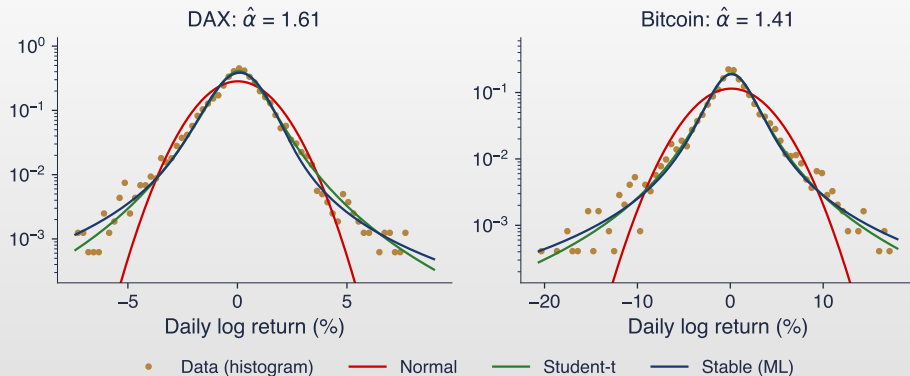
- $\hat{\alpha}$ well below 2 for all four series
 - ▶ S&P 500: $(2 - 1.540)/0.021 \approx 22$ standard errors from the Normal case
 - ▶ the heaviest tails: Bitcoin ($\hat{\alpha} = 1.410$)
- $\hat{\beta} < 0$ for the S&P 500 and the DAX: the left tail is heavier (crashes)
 - ▶ BET and Bitcoin: $\hat{\beta}$ within two standard errors of 0
- $\hat{\gamma}$: Bitcoin 1.54% vs S&P 500 0.58% a day; compare the sample standard deviations 3.49% and 1.21%
- McCulloch gives lower $\hat{\alpha}$ than ML for every series: the two methods weigh the tails differently

Fitted Densities on a Log Scale (1/2)



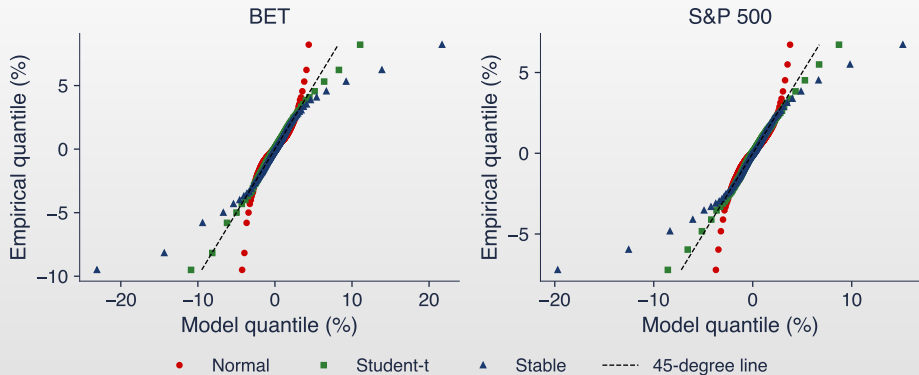
□ Normal (red): far too few large returns; Student-t (green) and stable (blue) both follow the body

Fitted Densities on a Log Scale (2/2)



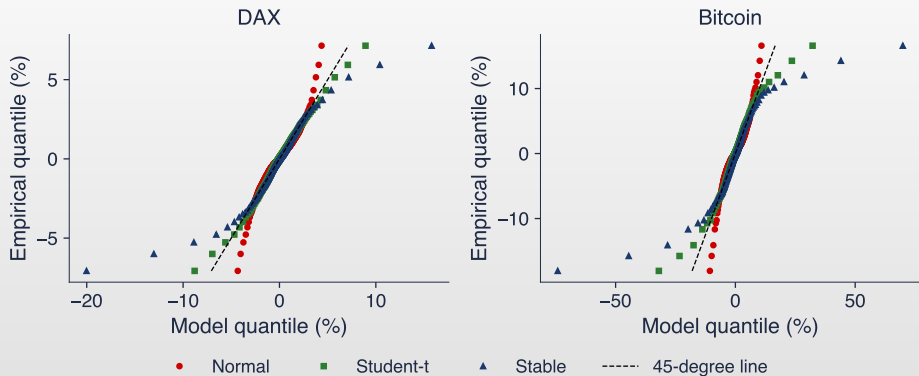
□ Far tails: the stable density stays above the data points; the Student-t is closer

QQ Plots against the Three Models (1/2)



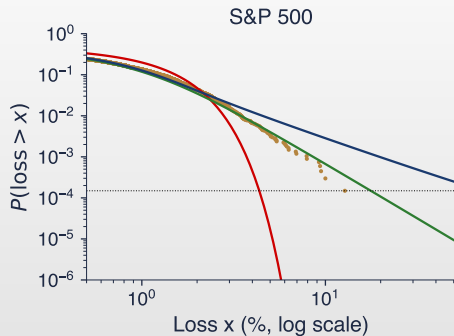
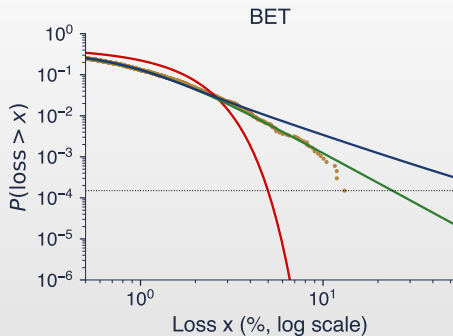
- A QQ (quantile–quantile) plot puts the empirical quantiles against the model quantiles; a good model lies on the 45-degree line

QQ Plots against the Three Models (2/2)



- Normal: too-short tails (steep ends); stable: too-long tails (flat ends, model quantiles far beyond the data); Student-t: the closest

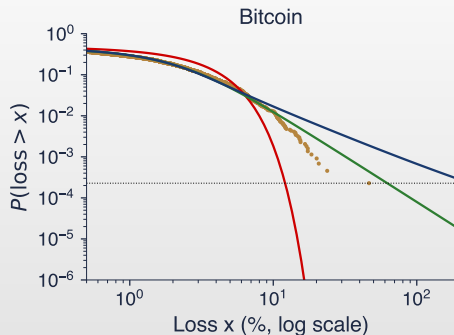
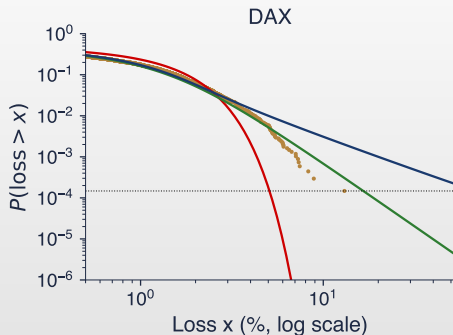
The Left Tail on Log-Log Axes (1/2)



• Data: share of days with loss $> x$ — Normal — Student-t — Stable one day in the sample

□ Points: share of days with a loss above x ; lines: the same probability under each fitted model

The Left Tail on Log-Log Axes (2/2)



• Data: share of days with loss $> x$ — Normal — Student-t — Stable one day in the sample

□ The empirical tail bends down faster than the stable line: the tail index of the data is larger than $\hat{\alpha}$

Counting Large Losses

	loss above	observed	Normal	Student-t	stable
S&P 500	5%	23	0.1	27.0	57.3
S&P 500	10%	1	≈ 0	4.5	19.1
S&P 500	15%	0	≈ 0	1.5	10.2
BET	10%	6	≈ 0	8.2	22.8
Bitcoin	10%	56	8.1	54.2	73.9
Bitcoin	20%	4	≈ 0	12.3	26.9

- Expected number of days = $n \times P(r < -x)$ under each fitted model
- Normal: almost no large losses; stable: too many, and the gap grows with the threshold; Student-t: closest to the counts
- S&P 500: 1 day below -10% in 6 714 days; the stable fit expects 19.1

VaR 1% and VaR 0.1% under the Three Models

	VaR 1% (%)				VaR 0.1% (%)			
	data	Normal	t	stable	data	Normal	t	stable
BET	4.1	3.2	4.1	4.9	9.5	4.3	10.9	23.1
S&P 500	3.5	2.8	3.5	4.5	7.2	3.7	8.6	19.7
DAX	4.2	3.3	4.0	5.0	7.1	4.3	8.8	20.0
Bitcoin	10.5	8.0	11.1	14.3	18.1	10.7	31.9	74.2

- VaR at level p : $\text{VaR}_p = -q_p$, the daily loss exceeded with probability p ([Chapter 10](#)); here p , because α is the stable index
- VaR 1%: the Normal model is too low, the stable model too high
- VaR 0.1%: the stable model gives roughly two to four times the empirical quantile

Which Model Fits Best?

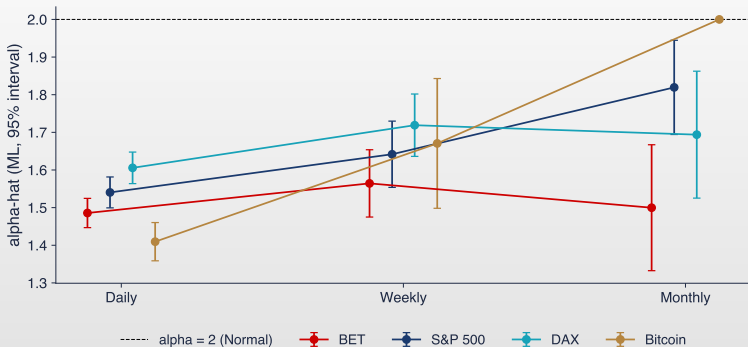
	AIC Normal	AIC Student-t	AIC stable	$\hat{\nu}$ (Student-t)
BET	23 407	20 996	21 088	2.43
S&P 500	21 650	19 702	19 794	2.67
DAX	23 901	22 595	22 697	3.09
Bitcoin	23 391	22 056	22 159	2.23

- AIC (Akaike information criterion) = $2k - 2\ell(\hat{\theta})$, with k the number of parameters; lower is better
- Both heavy-tailed models beat the Normal distribution by thousands of AIC points
- Student-t wins for every series, with fewer parameters; S&P 500: by 92 points
- $\hat{\nu}$ between 2 and 3.1: finite variance, but few finite moments beyond it

Recap: Fitting Real Returns

- ▣ All four series: $\hat{\alpha}$ between 1.410 and 1.606, far from 2
- ▣ The stable law fits the body much better than the Normal distribution
- ▣ It overstates the far tails: too many extreme losses, VaR 0.1% too high
- ▣ By AIC, the Student-t distribution fits better

Test 1: Does α Stay the Same When Returns Are Summed?

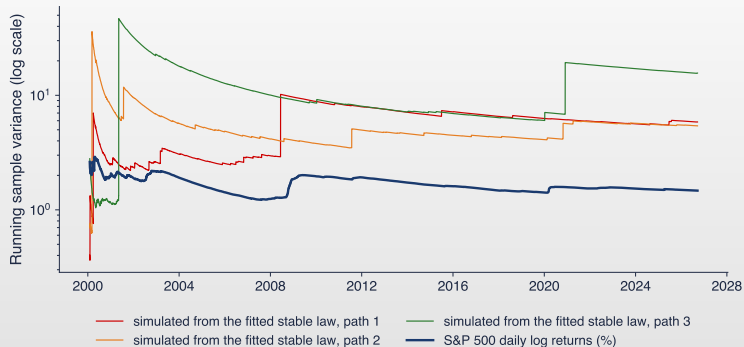


- If daily returns were i.i.d. stable, weekly and monthly sums would have the **same** α (stability)
- ML estimates: S&P 500 1.54 $\rightarrow$ 1.64 $\rightarrow$ 1.82; Bitcoin 1.41 $\rightarrow$ 1.67 $\rightarrow$ 2.00 (at the bound 2)

Reading the Aggregation Test

- ▣ $\hat{\alpha}$ rises with the horizon for the S&P 500, DAX and Bitcoin
 - ▶ DAX: $1.61 \rightarrow 1.72 \rightarrow 1.69$; the daily and monthly intervals do not overlap for the S&P 500
- ▣ BET: $1.49 \rightarrow 1.56 \rightarrow 1.50$, no clear rise
 - ▶ monthly samples are short (321 months): wide intervals
- ▣ The same pattern in the classic studies: Officer (1972), Akgiray & Booth (1988)
- ▣ A rising α is what the GCLT predicts for sums of finite-variance returns that converge slowly to the Normal distribution

Test 2: Does the Sample Variance Settle?



- S&P 500 (blue): after the jumps of 2008 and 2020 the running variance settles near 1.47
- Paths simulated from the fitted stable law, same length: final variances between 5.4 and 15.6, far above the data

Test 3: How Heavy Is the Far Tail?

- The stable α is driven by the whole distribution, mostly by the body
 - ▶ the tail index of the far tail alone is estimated with the Hill estimator [Hill \(1975\)](#) ([Chapter 5](#))
- For stock indices the far tail decays roughly like x^{-3} , the “inverse cubic law” [Gopikrishnan et al. \(1999\)](#)
 - ▶ a tail index near 3 means finite variance, and no stable law has it
- Our counts agree: the stable fits expect many more losses beyond 10% than observed
- [Lau, Lau & Wingender \(1990\)](#): further evidence against the stable model from the behaviour of sample moments

Why Do Daily Returns Look Stable?

- **Volatility clustering** (Chapters 2 and 9): calm and turbulent periods alternate
 - ▶ a mixture of Normal laws with changing variance has heavy tails and a finite variance [Blattberg & Gonedes \(1974\)](#)
 - ▶ fitted to one stable law, the mixture gives $\hat{\alpha} < 2$
- **Truncated and tempered stable laws**: stable at moderate scales, lighter tails far out
 - ▶ [Mantegna & Stanley \(1995\)](#) fitted a truncated Lévy flight to the S&P 500
 - ▶ [Grabchak & Samorodnitsky \(2010\)](#): such laws explain why returns look stable at short horizons and Normal at long ones

Verdict

- ▣ **Mandelbrot was right** about the facts
 - ▶ daily returns have heavy tails; the Normal distribution fails badly
- ▣ **The infinite variance is not supported**
 - ▶ $\hat{\alpha}$ rises with aggregation; the sample variance settles; the far tail is lighter than any stable tail
- ▣ Where stable laws remain useful
 - ▶ a benchmark for heavy tails; simulation; the scaling idea; data with truly infinite variance (some insurance losses, some crypto tokens)
 - ▶ the GCLT: a reminder that $\sqrt{n}$ -rules need a finite variance
- ▣ Next: [Chapter 5](#) estimates the tail index directly with extreme value theory

Recap: The Critique

- ▣ Stability predicts the same α at every horizon: the data show a rising α
- ▣ The sample variance of real returns settles; that of stable samples does not
- ▣ Volatility clustering and tempered tails explain the apparent $\alpha < 2$
- ▣ Heavy tails: yes; infinite variance: no

An Open Question

- **Have the tails of Bitcoin become lighter as the market matured?**
 - ▶ more participants, futures and ETFs (exchange-traded funds) since 2017–2024
 - ▶ if the tails got lighter, risk models fitted on old data overstate today's risk
- Why it is open: $\hat{\alpha}$ moves with volatility regimes, and yearly samples are short
- Full sample: $\hat{\alpha} = 1.410$ (SE 0.026), the lowest of our four series
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

How AI Could Help

- ▣ **Literature:** list papers on tail estimates for crypto assets and summarise their methods
- ▣ **Code:** draft a rolling-window McCulloch estimator with bootstrap intervals
- ▣ **Robustness:** propose other windows, ML instead of quantiles, Student-t ν instead of α
- ▣ **Explanation:** a first draft of the interpretation of a table of yearly estimates
- ▣ Example prompt
 - ▶ Write a Python function that estimates the stable alpha of daily log returns on rolling 365-day windows with McCulloch (1986), and adds 95% bootstrap intervals.

What to Check

- ▣ Parameterisation: S0 or S1; scipy uses S1 by default; the location differs by $\beta\gamma \tan \frac{\pi\alpha}{2}$
- ▣ Units: returns in % or in decimals; γ and δ change, α and β do not
- ▣ Overlapping windows are not independent: do not read the rolling estimates as separate tests
- ▣ Numbers: recompute one window with a second method (ML or the quantile tables)
- ▣ References: every cited paper must exist; check the DOI

Project Seed

- **Question:** is the tail of Bitcoin returns lighter now than in 2014–2018?
 - ▶ data: Bitcoin daily closes from the course data (EODHD), 2014–2026
- Steps
 - ▶ estimate α by McCulloch and by ML for each calendar year, with bootstrap intervals
 - ▶ repeat on returns divided by a rolling volatility: does the trend survive?
 - ▶ compare with the Student-t $\hat{\nu}$ of each year
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Formulas

Quantity	Formula
Stability	$aX_1 + bX_2 \stackrel{d}{=} cX + d, \quad c^\alpha = a^\alpha + b^\alpha$
Sum of n copies	$X_1 + \dots + X_n \stackrel{d}{=} n^{1/\alpha}X + d_n$
Characteristic function (S1)	$\exp\{-\gamma^\alpha t ^\alpha [1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2}] + i\delta_1 t\}$
S0 vs S1	$\delta_0 = \delta_1 + \beta\gamma \tan \frac{\pi\alpha}{2}$
Normal case	$S(2, \beta, \gamma, \delta) = N(\delta, 2\gamma^2)$
Tails	$P(X > x) \sim \gamma^\alpha c_\alpha (1 + \beta)x^{-\alpha}$
Moments	$E X ^p < \infty \iff p < \alpha \quad (\alpha < 2)$
CMS ($\beta = 0$)	$\frac{\sin(\alpha V)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1-\alpha)V)}{W} \right)^{(1-\alpha)/\alpha}$
McCulloch	$\nu_\alpha = \frac{q_{0.95} - q_{0.05}}{q_{0.75} - q_{0.25}}, \quad \nu_\beta = \frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$

Check Yourself

- **Question:** daily returns are i.i.d. $S(1.5, 0, \gamma, 0)$; by what factor does the scale grow from 1 day to 20 days?
 - ▶ **Answer:** $20^{1/1.5} = 7.37$, i.e. 1.65 times the square-root-of-time factor $\sqrt{20}$
- **Question:** a fitted law has $\alpha = 1.2$; do the mean and the variance exist?
 - ▶ **Answer:** the mean exists ($1 < 1.2$), the variance does not ($2 > 1.2$)
- **Question:** scipy reports $\delta = 0.05$ with $\alpha = 1.7$, $\beta = -0.2$, $\gamma = 0.6$; what is δ_0 ?
 - ▶ **Answer:** scipy uses S1, so $\delta_0 = 0.05 + (-0.2)(0.6) \tan(0.85\pi) = 0.05 + 0.061 = 0.111$
- Next: [Chapter 4](#), probability for finance

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